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Fair Business Models for Climate-Smart Agriculture: A Design Science Research Approach

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ABSTRACT

Organisations in agri-food value chains face urgent demands to address climate change, environmental degradation and social inequities through climate-smart agriculture (CSA). However, sustained CSA implementation remains constrained by business models (BMs) that fail to offer fair solutions to stakeholders. Existing business model innovation (BMI) literature lacks conceptualisation of fair BMs and methodology for operationalising fairness in BMs, leading to the research question: How can fair BMs for CSA be defined, designed and developed? Using design science research across five European case studies, the study defines a fair BM as a BM in which the design of the value proposition, creation, delivery and capture enables equitable value distribution, inclusive processes, transparency of information and respectful relationships among stakeholders. The study operationalises fairness across the BMI process with stakeholder engagement and change management, deriving eight design requirements and thirteen stakeholder-validated prototypes. The study advances BMI literature by offering prescriptive knowledge for operationalising fairness in agri-food BMI.

1 | Introduction

Organisations in agri-food value chains face mounting pressure to implement sustainable and regenerative practices that address climate change, environmental degradation and social inequities (Barrett and Gómez 2025). Among various approaches to transforming food systems, climate-smart agriculture (CSA) gained prominence for integrating farming practices that enhance productivity, build resilience to climate shocks and reduce greenhouse gas emissions (Erekalo et al. 2024; FAO 2013). Despite its potential, CSA remains constrained by current business models (BMs) in agri-food value chains that fail to provide solutions that are perceived as fair by stakeholders, undermining long-term commitment towards CSA (Rosenstock et al. 2020).

Fairness is understood as a concept of four dimensions encompassing equitable distribution of benefits and burdens (distributive), inclusive participation in decision-making (procedural), transparent and equitable access to information (informational) and respectful treatment of all stakeholders (interpersonal) (Gudbrandsdottir et al. 2021). These dimensions shape whether stakeholders perceive to be adequately rewarded, included, well-informed and treated with respect in their relationships, which are conditions necessary for sustained stakeholder engagement and long-term commitment to CSA (Ling et al. 2024; Rosenstock et al. 2020). When they perceive unfairness along one or more of these dimensions, stakeholders tend to disengage, reduce investments and resist implementation even when practices contribute to sustainability and

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regeneration outcomes (Prete and Samoggia 2023). Current BMs in agri-food value chains reinforce disengagement and resistance by structuring value creation, delivery and capture in ways that ignore one or more of the fairness dimensions. Achieving fairness in agri-food value chains when implementing CSA therefore requires business model innovation (BMI) towards fair BMs, through which stakeholders are deliberately engaged to embed fairness into BM configurations (Samoggia and Beyhan 2022). Farmers stand at the centre of this innovation, as they are the ones that take CSA implementation decisions and bear direct implementation costs, positioning farm BMs as the focal point for CSA-related innovation. Yet, farmers cannot innovate their BMs in isolation; fair BMs must be co-created with all relevant stakeholders to ensure legitimacy, effectiveness, contextual fit and value chain alignment (Isakhanyan et al. 2024).

The sustainable and regenerative BMI literature offer important starting points for BMI towards fair BMs but leaves significant gaps. Sustainable BMI addresses incorporating sustainability concerns in BM design (Bocken et al. 2014), while the emerging literature on regenerative BMs emphasises planetary health restoration and social well-being through activities promoting justice and fairness (Konietzko et al. 2023). However, neither stream of literature provides guidance on how to design fair BMs, particularly in multi-stakeholder agri-food contexts, and a conceptualisation of what constitutes a fair BM remains absent. Moreover, existing BMI frameworks (e.g., Geissdoerfer et al. (2017); Wirtz and Daiser (2018)) do not provide guidance for managing complex change processes when multiple stakeholders jointly want to develop fair BMs (Ulvenblad et al. 2018). These complex multi-stakeholder change processes underscore the need for studying and integrating change management, that is, the practice of applying a structured approach to managing business change, into BMI.

Sancak (2023) suggests that change management can increase the success rate of sustainable innovation through mechanisms such as clear change vision and active stakeholder engagement. By helping overcome resistance, securing stakeholder buy-in and building momentum for transformation (Hasan et al. 2025), change management can support farmers and their stakeholders in BMI towards fair BMs. However, the BMI literature shows limited attention to change management (Lang 2020). These knowledge gaps led us to pose the research question: *How can fair business models for climate-smart agriculture be defined, designed and developed?*

To answer the research question, we implemented a design science research approach (vom Brocke et al. 2020), suited for designing and evaluating artefacts through an iterative sequence of research activities that combined interviews, co-creation workshops and feedback sessions. The study derived eight design requirements (DRs) from stakeholder-identified fairness problems, then transformed DRs into thirteen fair BM prototypes and validated fair BM prototypes on relevance and plausibility.

The study makes several interrelated contributions to BMI literature. First, it proposes a definition and conceptualisation of a fair BM as a distinct concept. Second, the study proposes an integrated conceptual framework that combines the BMI process with change management elements, addressing the current limited guidance on the change process that can lead to fair BMs. Third, the study responds to calls for design-oriented approaches in sustainability transitions research by applying the integrated framework using design science research to generate prescriptive artefacts that illustrate how fairness can be operationalised in BMs. Fourth, the empirically based DRs and the portfolio of validated fair BM prototypes provide concrete mechanisms to address fairness during CSA implementation.

The paper is structured as follows. Section 2 elaborates the theoretical framework. Section 3 details DSR activities implemented across three stages, including case study contexts, data collection and analysis. Section 4 presents the results. Section 5 comprises the discussion, conclusions and implications.

2 | Theoretical Framework

2.1 | Conceptualising and Defining a Fair BM

A clear conceptualisation and definition of fair BMs is needed for BMI that address fairness. Clarifying what fairness in BMs means and what outcomes fair BMs can achieve is particularly important in agri-food contexts, where implementation of sustainable and regenerative practices require BMs that address inequities in risk-sharing, decision-making and benefit distribution (Rosenstock et al. 2020). In the literature, however, the conceptualisation of fair BMs is still in its infancy, and a clear definition remains absent. Mouchtaropoulou et al. (2024) and Oliveira et al. (2021), for instance, recognised fair BMs alongside sustainable BMs, referencing them as 'sustainable and fair business models', but do not provide a definition of what makes a BM fair. Recent work on regenerative BMs foregrounds 'justice and fairness', reframing BMs around ecological restoration and societal well-being, alongside multi-stakeholder value creation and multi-capital accounting (Konietzko et al. 2023). While being more explicit about fairness than the sustainable BM concept, the regenerative BM literature still lacks clear conceptualisation of what fairness means in BM.

The lack of conceptualisation of fairness in BM becomes salient in cases where a BM might be considered environmentally sustainable or regenerative but still be perceived as unfair (Schoneveld 2020). For example, a sustainable BM might achieve reduction in greenhouse gas emissions yet fail to qualify as fair if key stakeholders are excluded from decisions about how risks and rewards are allocated (Ponte 2022). Similarly, a regenerative BM may deliver net-positive ecological outcomes through soil restoration but still be perceived as unfair when the burden of implementation, monitoring and certification of the regeneration measures falls disproportionately on specific stakeholders (e.g., farmers) while other stakeholders capture most of the value (Schreefel et al. 2024). In such cases, perceived unfairness can undermine stakeholder trust, collaboration and willingness to invest even when BMs support practices that contribute to sustainability or regeneration outcomes (Zhou et al. 2020). These observations suggest that it is not enough for a BM to support sustainability or regeneration outcomes; it must also structure value creation, delivery and capture in ways that are perceived

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as fair by all stakeholders; otherwise, disadvantaged stakeholders resist or disengage. Fair BMs are therefore necessary to ensure long-term stakeholder commitment towards practices that contribute to sustainable and regenerative outcomes. The importance of fairness in supporting sustainability and regeneration motivates the conceptualisation of a fair BM as a distinct concept.

The study adapts the operationalisation of organisational fairness into the agri-food context by Gudbrandsdottir et al. (2021), which builds on Colquitt's (2001) four-dimensional framework of distributive, procedural, interpersonal and informational justice. Following Gudbrandsdottir et al. (2021), the study extends this operationalisation to the BM level, positioning the four fairness dimensions as criteria for BM design. This operationalisation establishes the distinctiveness of fair BMs relative to sustainable and regenerative BMs (see Table 1). Sustainable BMs, including variants such as inclusive BMs (Schoneveld 2020) and stakeholder-oriented BMs (Freudenreich et al. 2019), may attend to aspects of fairness through participation of income-constrained groups or mutual value creation, but they treat fairness as implicit in broader sustainability aims rather than explicit as a design criterion. Regenerative BMs similarly acknowledge fairness but tend to keep it implicit as part of broad justice narratives (Konietzko et al. 2023), without systematically operationalising fairness into BM design. In contrast, the fair BM conceptualisation developed in this study explicitly takes fairness as the central design lens.

Drawing parallels from conceptualisations and definitions of sustainable and regenerative BMs, we propose the following definition of a fair BM: *A fair BM is a business model in which the design of the value proposition, value creation and delivery and value capture enables equitable value distribution, inclusive processes, transparency of information and respectful relationships among stakeholders*. By explicitly addressing the four fairness dimensions (Gudbrandsdottir et al. 2021) in BM design and development, a fair BM provides fair value to stakeholders in a way that:

- All stakeholders receive fair compensation and benefits reflecting their labour, investment, risk-taking and contribution to innovation (distributive fairness).
- All stakeholders have a choice and opportunity to engage in relevant decision-making processes (procedural fairness).
- All stakeholders have access to high-quality decision-relevant information (informational fairness).
- All stakeholder interactions are characterised by dignity and respect (interpersonal fairness).

Stakeholders' perceptions of fairness are inherently subjective; what is perceived as fair depends on individual preferences, values, interests and positions within the value chain (Oyedijo et al. 2021; Sun et al. 2021). Although a fair BM would ideally address all four dimensions, the importance of one or the other may vary depending on the context. For example, when a BM supports practices requiring substantial investment, farmers may face a higher challenge regarding distributive fairness than regarding interpersonal fairness. However, the four dimensions are not independent: 'although separate dimensions, they affect each other' (Gudbrandsdottir et al. 2021, 7). Even when distributive fairness is the most pressing concern, how farmers are treated in negotiations and interactions shape how they perceive the fairness of the outcomes they receive. Similarly, information sharing and information quality between business partners are antecedents of procedural fairness (Gudbrandsdottir et al. 2021). Hence, the study proposes that fair BMs should be designed and developed to address the most pressing fairness concerns in response to contextual priorities, but a BM that is unfair on any one dimension is not a fair BM. This interrelatedness helps explain why BMs that achieve sustainability or regeneration outcomes can still be perceived as unfair when their design neglects or undermines any of the fairness dimensions. Thus, a BM that achieves greenhouse gas emission reduction but undermines procedural fairness by increasing administrative burden cannot be considered a fair BM.

2.2 | BMI Process and Change Management

Due to the subjectivity of fairness perceptions by upstream and downstream agri-food value chain actors co-shaping BM configurations, multi-stakeholder engagement is important in fair BMI for CSA. The BMI literature presents examples of BMI process frameworks (e.g., Frankenberger et al. 2013;

TABLE 1 | Conceptualisation of fair and other business models.

Type of BM	Treatment of fairness	References
Sustainable BM	Fairness is implicit in sustainability aims; not explicitly operationalised in BM design	Bocken et al. (2014); Geissdoerfer et al. (2018); Lüdeke-Freund et al. (2024); Schaltegger et al. (2015); Stubbs and Cocklin (2008)
Regenerative BM	Fairness is part of justice oriented, net-positive socio-ecological transformation; not explicit on what is fairness and how it is operationalised in BM design	Drupsteen and Wakkee (2024); Konietzko et al. (2023)
Fair BM	Fairness is the central design lens; design of BM components are explicitly oriented towards enabling distributive, procedural, informational and interpersonal fairness outcomes	Colquitt (2001); Gudbrandsdottir et al. (2021); Prete et al. (2025); Samoggia and Beyhan (2022)

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Geissdoerfer et al. 2017; Lindgardt et al. 2015; Wirtz and Daiser 2018), but these frameworks do not explicitly address fairness. Nevertheless, the 4I framework of Frankenberger et al. (2013) provides an important starting point for our study because it explicitly recognises the various stakeholders in the ecosystem of the innovating organisation. In the 4I framework, the innovating organisation and stakeholders move from recognising opportunities and challenges (initiation), to exploring alternative solutions (ideation), to combining and refining them into coherent BM designs (integration) and finally to putting BM design into practice (implementation). However, stakeholder engagement alone is insufficient because fair BMI requires actors to abandon familiar practices and to commit to substantial organisational and behavioural changes (Barth et al. 2021; Ulvenblad et al. 2018), which require change processes that existing BMI frameworks do not explicitly address.

Change management theories can complement existing BMI frameworks. Lewin's (1951) classical three-stage model of unfreezing, change and refreezing has provided an important theoretical foundation, whereas contemporary approaches such as those of Mento et al. (2002) and Kotter (2007) have extended rather than replaced its core logic by dividing the stages into more detailed steps (Errida and Lotfi 2021). Although the model has been criticised as overly linear, Burnes (2004, 2020) demonstrates that Lewin conceived the stages as iterative and participative processes highlighting that change should be driven by those affected by it, making it particularly suited to multi-stakeholder co-creation processes such as fair BMI. Each stage addresses the distinct requirements for fair BMI. *Unfreezing* creates awareness of limitations of existing BMs, building urgency for BMI and fostering willingness to innovate. *Change* introduces new ideas and behaviours, typically characterised by uncertainty, learning and gradual adaptation as stakeholders develop competence with new approaches. *Refreezing* institutionalises new practices into routine operations, though it requires careful monitoring to avoid locking-in configurations. Wirtz and Daiser (2018) emphasise the relevance of change management and apply it only to the implementation phase of BMI. In the agri-food context, successful CSA implementation requires coordinated action and sustained commitment from stakeholders with varying interests, capabilities and power positions (Isakhanyan et al. 2024); change management must therefore be integrated across all the 4I phases of initiation, ideation, integration and implementation.

2.3 | An Integrated Framework for Fair BMI

This study proposes an integrated conceptual framework for fair BMI to guide the design and development of fair BMs in science and practice (Figure 1). The framework integrates the 4I framework of Frankenberger et al. (2013) with the three-stage change model of Lewin (1951). In this framework, innovative activities, stakeholder engagement and change management principles are coordinated to operationalise fairness throughout the BMI process. The new framework consists of four stages, which align the four stages of the 4I framework with the three stages of Lewin (1951). **Initiation** is aligned with unfreezing where awareness of fairness limitations in current BMs is built and readiness for change is established; **ideation** spans the boundary between unfreezing and change as challenging existing assumptions and exploring alternatives are intertwined in multi-stakeholder settings; **integration** is aligned with change, where new approaches addressing fairness are developed and validated; and **implementation** is aligned with refreezing, where fair BMs are institutionalised.

Figure 1 illustrates the activities per change management stage that are needed in developing fair BMs for CSA. The activities comprise the change management principles that guide change processes; the stakeholder engagement that applies change principles in stakeholder interactions; and the BMI activities discussed for each stage. The nature of stakeholder engagement evolves along the different stages. In the **initiation stage**, engagement is consultative: Stakeholders

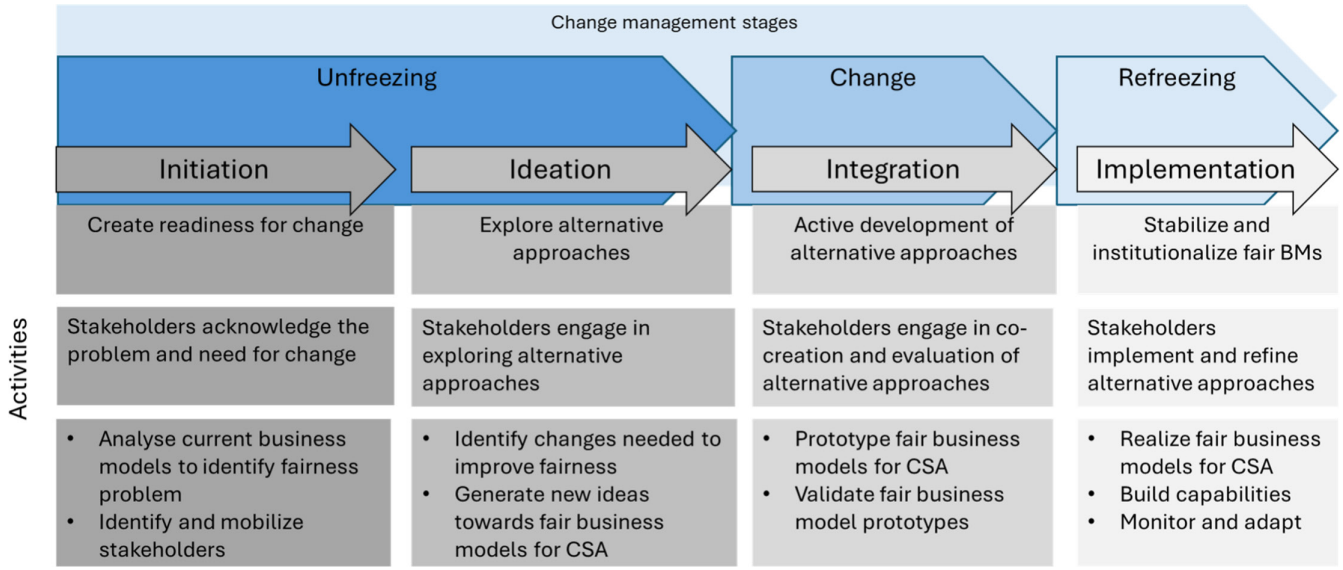


FIGURE 1 | Integrated framework for fair business model innovation for climate smart agriculture.

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build collective acknowledgement of challenges related to fairness and create readiness for change by analysing current BM performance and identifying those stakeholders who can drive the change process. In the **ideation stage**, engagement becomes explorative: Stakeholders explore potential solutions to address fairness in the BM, bringing diverse perspectives and expertise to collaborative innovation processes. In the **integration stage**, engagement is co-creative: Stakeholders co-create and refine fair BM design by elaborating detailed prototypes, validating prototype relevance for addressing fairness and identifying implementation requirements and success factors. In the **implementation stage**, engagement is evaluative: Stakeholders institutionalise the fair BM through pilot implementation, monitoring and evaluation of fairness outcomes and business performance, continuous adaptation and improvement as well as systematic documentation of lessons learned.

The integrated framework offers a coherent structure of a process for designing and developing fair BMs by aligning BMI activities, stakeholder engagement and attention to change management throughout all stages of the BMI process.

3 | Methodology: Design Science Research

The integrated framework (Figure 1) provides the overall logic for our empirical study and was applied and tested following a design science research approach, suited for designing artefacts that solve real-world problems (Romme and Reymen 2018; vom Brocke et al. 2020). In design science research, artefacts are understood as purposeful constructs, models and methods that provide prescriptive knowledge about how to act in a given domain (Seckler et al. 2021). In this study, the artefacts are the fair BM prototypes, offering farmers and their stakeholders concrete templates on how to operationalise fairness in their BM. Design science research involves six activities: problem identification; objective definition; design and development; demonstration; evaluation; and communication (vom Brocke et al. 2020). These activities often require triangulation of multiple research methods (Johannesson and Perjons 2021). Since the conceptualisation and operationalisation of fairness in BMs are still in its infancy, the study relied on exploratory qualitative methods, specifically semi-structured interviews and workshops, to gather stakeholder perspectives and generate design knowledge in a domain where literature provides limited guidance (Seckler et al. 2021). Figure 2 illustrates how we implemented the integrated framework for fair BMI through the design science research activities.

The initiation and ideation stages were operationalised through the *problem identification and objective definition* activities respectively to derive DRs. DRs are generic requirements that define what an artefact must achieve or enable (Moder et al. 2025). In our study, the DRs define what BMI must enable to address fairness in CSA implementation. The integration stage was operationalised through the *design and development* activities, which transformed the DRs into fair BM prototypes using BM patterns in the literature as building blocks consistent with design science principles that established typologies and taxonomies can inform artefact design (Lüdeke-Freund et al. 2018; Seckler et al. 2021), and through the *demonstration and evaluation* activities, where stakeholders validated the prototypes' relevance and plausibility. The implementation stage of the framework, which involves the actual institutionalisation of fair BMs into routine operations, was beyond the scope of this study. Central and unique to the execution of the activities were the co-creation processes defined as a participatory, open, active and creative process to design and decision-making that involves multiple stakeholders (Ind and Coates 2013; Jones 2018).

The design science research activities were conducted across five case studies as part of a Horizon Europe project on behavioural change towards CSA. The case studies were selected to represent the diverse European farming systems and food products: pig

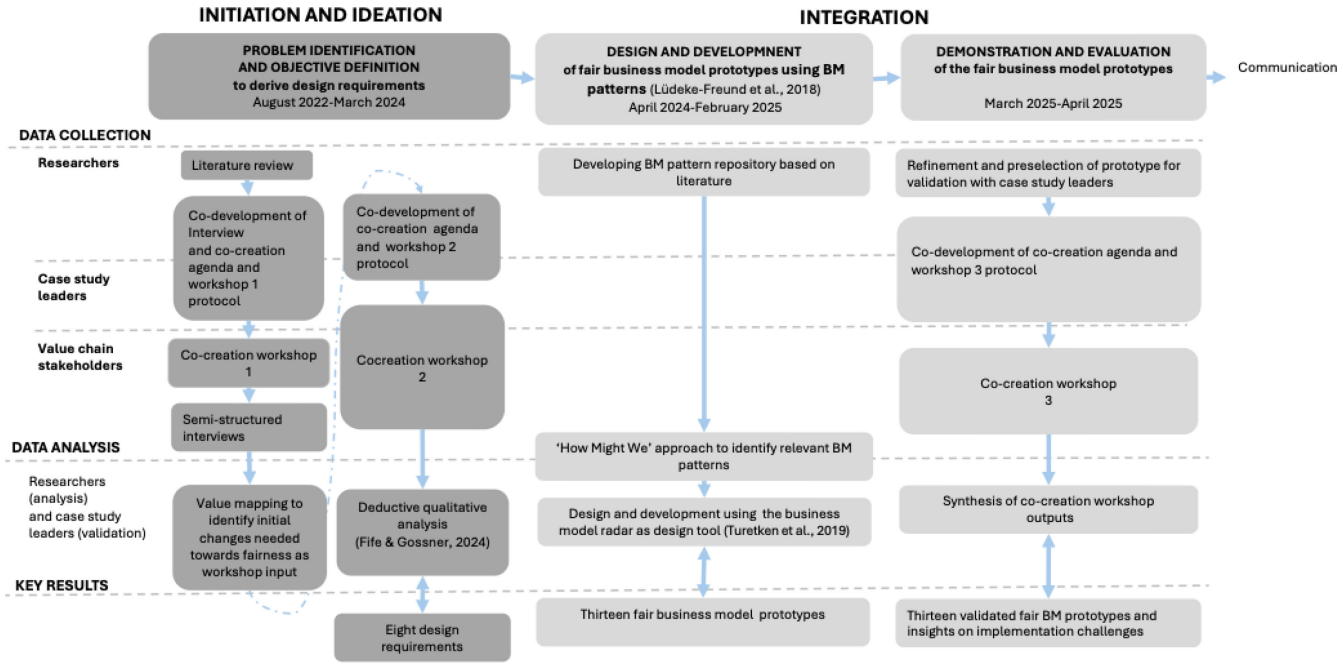


FIGURE 2 | Design science research approach.

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production in Denmark; dairy production in Germany; wheat production in Lithuania; potato and onion production in the Netherlands; and apple production in Spain. These five farming systems account for approximately 45% of EU agricultural land and the majority of EU exports, covering various CSA practices within diverse institutional contexts. This diversity enabled the identification of both context-specific and cross-cutting fairness problems that can guide the design and development of broadly applicable fair BM solutions.

As shown in Figure 2, the activities were conducted as an iterative collaboration between three main stakeholder groups: researchers, case study leaders and various agri-food value chain stakeholders. Case study leaders were (intermediary) organisations familiar with the local farming systems and value chains. They acted as informants to support data collection by conducting interviews in local languages, recruiting workshop and interview participants, facilitating workshop organisation and providing feedback and translation of the materials. The main interview respondents were the value chain stakeholders, including farmers, processors, retailers, advisors, technology providers, policymakers and consumer representatives. The recruitment of the value chain stakeholders through purposive sampling was supported by the case study leaders through their professional networks. Three criteria guided the recruitment: (1) representation across key value chain segments; (2) direct involvement in or influence over CSA implementation decisions; and (3) ability to represent the perspectives of their respective stakeholder group. Throughout the design science research process, the three stakeholder groups interacted iteratively: Preliminary insights (DRs and prototypes) were co-developed with case study leaders, presented to value chain stakeholders in co-creation workshops and subsequently refined by the research team based on stakeholder feedback. Whereas participant continuity across the three workshop rounds was intended, stakeholder availability over the research period limited continuation. Continuity in the process was maintained through the case study leaders, who participated in all rounds as recruiters, facilitators, translators and co-creators of workshop design and preliminary insights and through a recap of previous sessions at the start of subsequent workshops. The case study leaders also ensured consistency across the activities conducted in local languages. Prior to the conduct of the workshops, case study leaders validated pre-translated workshop materials provided by the researchers using DeepLPro and attended training sessions. Case study leaders also translated workshop outputs into English, which were reviewed by researchers.

3.1 | Data Collection

In *problem identification and objective definition*, we collected data through semi-structured interviews and two rounds of co-creation workshops in the five case studies to understand the fairness problem when implementing CSA in agri-food value chains and what changes towards fairness stakeholders considered necessary. An initial round of co-creation workshops bringing together 80 participants was organised in 2023 in the five case studies to familiarise stakeholders with the BMI process to create readiness for change, as prescribed in the Initiation phase. In parallel, 69 semi-structured interviews with stakeholders across the five case studies were conducted in January–March 2023 to explore current CSA practices, challenges and opportunities and how existing BMs support or hinder CSA. Insights from these interviews provided input for the second round of co-creation workshops held in February and March 2024. These workshops brought together 58 participants representing a range of value chain stakeholders (see Table 2) to discuss what fairness means in their contexts, co-identify and prioritise changes needed in BMs to address fairness and indicate which actors could drive the co-prioritised changes. To mitigate concerns around

TABLE 2 | Overview of the five co-creation workshops held in 2024.

Case	Date and place/format	Value chain stakeholders represented	No. of participants Total: 58
Wheat farming, Lithuania	13/3/2024, Vilnius/hybrid	Farmer (2), farmer association (1), technology provider (1) retailer (1), policymaker (1)	6
Dairy farming, Germany	29/02/2024, Heissenhoflnzell/hybrid	Farmer (2), breeding association (1), dairy processing (2), feed producer (1), retailer/trader (2), agricultural advisor (2), policy/government officer (2), consumer (2)	14
Pig farming, Denmark	8/3/2024, Herning/in-person	Farmer (1), feed supplier (2), researcher (1), policymaker (2), technology provider (7), advisor (1), business developer (1), investor (1)	16
Organic apple farming, Spain	5/3/2024, Pamplona/in-person	Farmer/processor (juice) (3), policymaker (1), researcher (2), advisor (3)	9
Onion and potato farming, the Netherlands	20/03/2024, Sint Annaland/in-person	Seed producer (1), farmer (2), potato, breeder (1), supplier (1), advisor (3), policymaker (2), processor (1), researcher (1), certifier (1)	13

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power asymmetries, fairness concerns and proposed changes were first collected individually in writing from participants before any plenary discussion. Prioritisation then used structured dot voting to ensure that each stakeholder type had an equal say. The workshop protocol and materials for the second co-creation workshop, detailing these procedures, are provided in Supporting Information A.1. The workshop results were used to develop DRs.

In *design and development*, we transformed the DRs derived from stakeholder-identified changes into BM prototypes drawing on the approach developed by Lüdeke-Freund et al. (2018). BM patterns are recurring configurations of BM components that can be (re)used as design building blocks for new BMs (Lüdeke-Freund et al. 2018; Remane et al. 2017). The approach was used because stakeholders, while clearly articulating changes needed towards fair BMs, had limited knowledge and experience in translating these changes into new BMs. To find suitable BM patterns, we conducted a literature study. Based on 25 articles (see Appendix B), we developed a repository of BM patterns that could help transform DRs into fair BM prototypes.

In *demonstration and evaluation*, we asked stakeholders to assess the fair BM prototypes and validate the created artefacts during the third round of co-creation workshops with the five case studies (Dimov et al. 2023; vom Brocke et al. 2020). Between March and April 2025, five co-creation workshops were conducted, involving 75 stakeholders (Table 3). The workshop protocol and materials for the third co-creation workshop are provided in Supporting Information A.2.

A subset of prototypes was pre-selected for validation based on discussions between researchers and case study leaders, considering the case study context that includes confirmed participant attendance, to ensure each prototype received thorough assessment. Prototypes and their assumptions were provided in advance as preparatory material reducing information asymmetries between participants. During the workshop, stakeholders worked in small groups of three to five participants, organised by case study leaders based on area of expertise and type of representation in the value chain, to review and validate fair BM prototypes. Stakeholders evaluated the relevance and plausibility of the key BM assumptions of the prototypes and collectively mapped assessments onto an adapted Assumption Prioritisation Canvas (Huryn 2024). *Relevance* was assessed based on the prompt 'Does this assumption matter?', capturing whether stakeholders consider the assumption important for the prototype to function as intended, and *plausibility* based on the prompt 'Does this assumption sound realistic? Have I seen this happen before?', capturing whether stakeholders consider the mechanism as realistic based on observed or experienced precedents. In addition, stakeholders identified implementation challenges, including practical barriers, coordination requirements and contextual constraints.

3.2 | Data Analysis

Data analysis aimed to (1) derive DRs for BMI from stakeholder-identified changes towards fair BMs, (2) translate the DRs into fair BM prototypes using BM patterns and (3) synthesise stakeholder validation of the prototypes. Each analytical step used the outputs of the preceding activities as inputs.

Consistent with the coding procedure in deductive qualitative analysis (Fife and Gossner 2024), the first author mapped the 33 stakeholder co-identified changes from the co-creation

TABLE 3 | Overview of the co-creation workshops held in 2025.

Case	Date and place	Fair BM prototypes validated	Value chain stakeholders represented	No. of participants Total: 75
Wheat farming, Lithuania	6/03/2025, Vilnius	Open source for CSA, transparency branding, consumer engagement platform, peer-to-peer knowledge sharing	Farmer (4), farmer association (1), technology provider (1), policymaker/ government (5), consumer organisation (1) advisor (4)	16
Dairy farming, Germany	14/3/2025, farm in Grabenstätt	Pay-for-success, consumer-driven pricing, bio-districts, data coopetition network	Farmer (3), breeding association (1), dairy processing (2), advisor (5), policy/government (2)	13
Pig farming, Denmark	7/3/2025, SKOV A/S	Pay-for-success, risk-sharing fund, retail choice editing	Farmer (1), policymaker (1), technology provider (4), advisor (4), vocational training (1)	11
Organic apple farming, Spain	1/4/2025, INTIA	Peer-to-peer knowledge sharing, crowdfarming, shared resource pool	Farmer/processor (1), farmer (8), advisor (4), consumer org (2), marketing association (1), commercial advisor (1)	17
Onion and potato farming, the Netherlands	12/3/2025, Klaaswaal	Pay-for-success, risk-sharing fund, retail choice editing, outcomes-based pricing	Farmer (5), auctioneer (1), advisor (4), trader (6), certifier (1), consumer organisation (1)	18

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workshops using the four fairness dimensions as sensitising categories to map and cluster the co-identified changes (see Appendix A). A DR was derived from each cluster, defining what BMI must enable to address fairness. The mapping, clustering and DRs were reviewed by two co-authors, with minor disagreements resolved through discussion and iterative refinement. Inter-coder reliability was not calculated. The DRs were presented and discussed with case study leaders to review clarity and contextual relevance and to confirm that they remain aligned with stakeholder co-identified changes. This analytical process supported rigour and enacted the co-creation logic of the DSR approach, such that DRs emerged as co-created artefacts. Consequently, the DRs together with the BM patterns became the main inputs for the design and development of fair BM prototypes.

For each DR, we formulated a 'How might we' question, framing it as a design challenge (e.g., DR1 was transformed into 'How might we fairly incentivize farmers to implement CSA?') to guide the search for adaptable and suitable BM patterns in the repository that could be used to translate the DRs into fair BM configurations. The 'How might we' approach is a technique used in design thinking and design science research for transforming an initial problem definition into a suitable design challenge (Asano 2023). Selected BM patterns informed the design configurations of the fair BM prototypes, which we structured using the business model radar by Turetken et al. (2019) (Figure 3).

The Radar was chosen over traditional design tools like BM canvas because it depicts BMs through a network-centric approach and represents value creation as a collaborative multi-stakeholder process, aligning with our conceptualisation of fairness. In designing and developing each prototype, the farmer was taken as the focal actor, and stakeholders were added as identified in the case studies. For each prototype, we defined the role of every stakeholder identified by specifying the co-production activities and possible costs and benefits. The prototypes developed were presented and discussed with case study leaders to ensure contextual relevance.

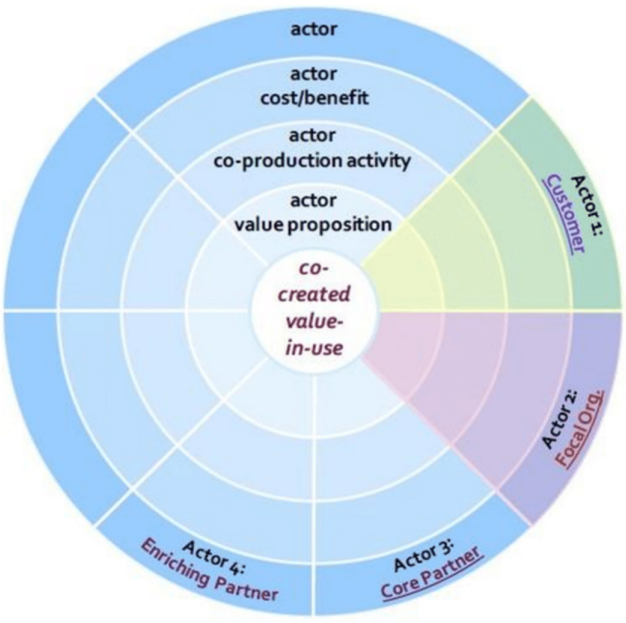


FIGURE 3 | The business model radar (Turetken et al. 2019).

In the analysis of the demonstration and evaluation activities outputs, prototype-level ratings were derived from the assumption-level assessments, reached through group consensus. Because the canvas is a mapping tool, stakeholders positioned each assumption along a continuum; assumptions mapped between the high and low poles were interpreted as mid, reflecting partial familiarity with analogous precedents or mixed assessments. This mid-category was emergent from stakeholder mapping behaviour rather than pre-defined in the workshop protocol. The implementation challenges stakeholders identified during the discussions were used to corroborate the assumption-level assessments, particularly assessments mapped at the boundaries between high and low ratings. Because the prototypes represent BMI that have not yet been implemented, the combination of the (1) assumption level assessments, which captures stakeholders' judgement of whether assumptions are realistic grounded in observed mechanism in practice, and (2) implementation challenges, which capture specific institutional barriers and missing enabling conditions, provide the strongest available basis for assessing plausibility. Low plausibility therefore indicates limited experience or familiarity with prototype mechanisms and significant implementation barriers in the cases studied, not infeasibility. Results were organised per prototype and compared across the five case studies to identify recurring patterns and contextual differences.

4 | Results

4.1 | DRs

As a result of the iterative approach, we identified eight DRs for BMI. Each DR captures what stakeholders believe BMI must accomplish to address fairness in agri-food value chains when implementing CSA. The DRs address distributive, procedural and informational fairness. The absence of interpersonal fairness as a design priority for BMI is discussed in Section 5.2.

4.1.1 | DR1: Fair Incentives for Farmers

Stakeholders from the Netherlands, Denmark and Germany emphasised the need for economic incentives that 'fairly reward farmers' for implementing CSA practices. This DR translates the policy-level desire for economic support directing BMI towards creating value capture arrangements that fairly reward farmers for CSA implementation, such as premium pricing systems, risk-sharing mechanisms and long-term contracts that recognise CSA investments.

4.1.2 | DR2: Equitable Distribution of Benefits and Costs

Stakeholders from Lithuania, the Netherlands, Germany and Denmark recognised that current value distribution mechanisms often disadvantage certain actors, primarily the producers highlighting the need for 'fair distribution of margins and

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financial support'. This DR guides the BMI process into redesigning value capture mechanisms to ensure farmers receive a fair share of sustainability premiums and that implementation costs are distributed proportionally across the value chain.

4.1.3 | DR3: Facilitation (Not Regulation) Towards Sustainability

Stakeholders from all case studies highlighted the need for 'more facilitation' rather than purely regulatory approaches to encourage CSA transitions. This DR guides BMI towards value creation and delivery mechanisms, such as technical assistance and cross-sectoral collaborations.

4.1.4 | DR4: Engagement and Collaboration Among Stakeholders

Stakeholders from Spain, Lithuania and Denmark emphasised the importance of collaborative relationships and meaningful engagement among value chain actors. This DR guides the development of mechanisms within BMs that enable meaningful participation from all value chain actors, particularly those with less power.

4.1.5 | DR5: Improved Access to Technology

Stakeholders from Lithuania and Spain highlighted the need for 'increased access to appropriate technologies'. This DR directs BMI towards designing mechanisms that can facilitate access to technologies essential for CSA implementation, such as shared technology platforms.

4.1.6 | DR6: Transparency Among Value Chain Actors

Stakeholders from the Netherlands and Denmark emphasised the need to 'make the value chain transparent'. This DR guides the BMI towards creating transparency mechanisms that make relevant information accessible and useable for all stakeholders, particularly those with limited resources.

4.1.7 | DR7: Consumer Awareness and Knowledge About Sustainable Choices

Stakeholders from Spain, Germany and Lithuania emphasised the importance of 'raising awareness among consumers about importance and impact of buying sustainable products'. This DR directs BMI towards creating meaningful connections between producers and consumers, providing decision-relevant information and enabling informed purchasing choices.

4.1.8 | DR8: Exchange of Sufficient and Reliable Information

Beyond transparency and consumer awareness, stakeholders from Spain, Germany and Denmark identified the broader need for sufficient and reliable information exchange throughout the value chain. This DR guides the BMI towards facilitating exchange of reliable, timely and relevant information through improved information exchange infrastructure.

4.2 | The Fair BM Prototypes

Fair BM prototypes represent artefacts that illustrate how fairness can be operationalised in BMI (Table 4). Consistent with the pattern-based approach of Lüdeke-Freund et al. (2018), the prototypes build on established BM patterns. The contribution lies in systematically reconfiguring existing patterns to embed fairness mechanisms across BM components. Prototypes such as the pay-for-success, crowdfarming, bio-districts, choice editing and open source adapt well-known patterns, whereas the rest present more original configurations. The detailed descriptions of the fair BM configurations are presented in Supporting Information B.

Six fair BM prototypes primarily address *distributive fairness* creating fair incentives for farmers (DR1), equitable distribution of costs and benefits (DR2) and improved access to technology (DR5). **Pay-for-success** restructures how CSA value is captured through outcome-based compensation systems tied to measurable CSA outcomes such as soil health improvements or carbon reduction. **Crowdfarming** restructures value capture through direct consumer–farmer adoption models that provide upfront financing and guaranteed markets for CSA products. **Consumer-driven fair pricing** and **risk-sharing fund** redistribute costs and benefits more equitably, the first by empowering consumers and farmers and the second by pooling risk across multiple stakeholders when adopting CSA practices. **Outcomes-based pricing** ties payments to measurable performance outcomes, and **shared resource pool** enables access to shared machinery and tools, thereby lowering barriers to technology access for farmers. Across these prototypes, procedural fairness mechanisms are embedded in the value creation and delivery. For instance, the risk-sharing fund creates and delivers value through collaborative fund management and the shared resource pool through collective decision-making on equipment access.

Three prototypes primarily address *procedural fairness* operationalising fairness through facilitative support (DR3), multi-stakeholder engagement and collaboration (DR4) complemented by distributive fairness where farmers' risk is reduced. **Bio-districts** and **retail choice editing** create supportive institutional and market environments for sustainable and regenerative practices. Bio-districts accomplishes this through territorial agreements where farmers, citizens and public authorities formally commit to sustainable resource management based on CSA principles. Retail choice editing implies prioritising sustainable options while 'editing out' unsustainable alternatives in retail shelves. **Open source for CSA** enables collaboration of stakeholders by using digital platforms for community-based knowledge sharing, collaborative research and joint innovation. The prototypes focus on structuring inclusive participation and engagement processes.

Four prototypes address *informational fairness* operationalising fairness through transparency among actors (DR6), consumer awareness and knowledge about sustainable choices (DR7) and

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TABLE 4 | Operationalisation of fairness in design requirements and fair business model prototypes.

Initiation and ideation		Integration			
Design requirements	**Fairness dimensions**	**BM patterns (references)**	**Fair BM prototypes**	**Fair BM configuration**	**Key actors**
DR1. Fair incentives for farmers	Distributive, procedural	Pay-for-success (Schroedel 2023); adopt a stewardship role (Ulvenblad et al. 2019); results model (Pieroni et al. 2020)	Pay-for-success	Farmers are compensated through outcome-based contracts tied to measurable CSA results, aligning rewards with verifiable sustainability performance	Farmers, investors, local government, certifiers
		Crowdfarming (Miranda et al. 2023)	Crowdfarming	Farmers set prices and receive upfront financing through direct consumer adoption models, reducing market dependency	Farmers, consumers, platform facilitator
DR2. Equitable distribution of benefits and costs	Distributive, procedural	Differential pricing (Schroedel 2023); pay what you want (Remane et al. 2017); price set by users (Curtis 2021)	Consumer-driven Fair Pricing	Farmers and consumers jointly determine prices through transparent mechanisms that account for sustainability costs and fair compensation	Farmers, consumers, retailers
		Risk-sharing (Remane et al. 2017); Revenue sharing (Schirmer et al. 2021)	Risk-sharing fund	Farmers are protected from income losses during transition to CSA implementation through collaborative funds where multiple stakeholders share risk	Farmers, public sector, private sector, insurance providers
DR3. Facilitation (not regulation) towards sustainability	Procedural, distributive	Bio-districts (Miranda et al. 2023)	Bio-districts	Farmers participate in territorial governance agreements that create supportive policy and market environments for CSA implementation	Farmers, citizens, public authorities, tourism firms
		Choice editing by retailers (Bocken et al. 2014)	Retail choice editing	Farmers gain guaranteed shelf space and marketing support as retailers curate product selections to prioritise sustainable options	Farmers, retailers, consumers
DR4. Engagement and collaboration among stakeholders	Procedural, informational	Open source (Curtis 2021); collaborative community platforms (Schroedel 2023)	Open source for CSA	Farmers participate as equal contributors alongside innovators and researchers in community-driven knowledge sharing and joint innovation	Farmers, innovators, research institutions, sponsors

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TABLE 4 | (Continued)

Initiation and ideation		Integration			
Design requirements	**Fairness dimensions**	**BM patterns (references)**	**Fair BM prototypes**	**Fair BM configuration**	**Key actors**
DR5. Improved access to technology	Distributive, procedural	Performance-based contracting (Kralewski 2016); performance or results model (Pieroni et al. 2020)	Outcomes-based pricing	Farmers access technology without upfront costs, paying only based on measurable performance outcomes, sharing implementation risks with providers	Farmers, technology providers, third-party evaluators
		Shared resource (Schroedel 2023); Pooling System (Pieroni et al. 2020); Rent Instead of Buy (Remane et al. 2017)	Shared resource pool	Farmers access shared machinery and tools through cooperative platforms on rental or pay-per-use basis, reducing individual ownership costs while enabling collective decision-making	Farmers, cooperatives, equipment providers, platform operators
DR6. Transparency among value chain actors	Informational, procedural	Transparency branding (Böttcher et al. 2024); radical transparency (Schroedel 2023)	Transparency branding	Farmers and supply chain partners disclose verified information about environmental impacts and production processes, building consumer trust	Farmers, processors, certifiers, retailers, consumers
DR7. Consumer awareness and knowledge about sustainable choices	Informational, procedural	Behaviour change; encourage sufficiency (Schroedel 2023)	Consumer engagement platform	Farmers benefit from consumer behaviour shifts driven by digital tools that provide insights into environmental impacts and personalised recommendations	Farmers, consumers, retailers, technology providers
DR8. Exchange of sufficient and reliable information	Informational, procedural	Peer-to-peer space sharing platforms (Curtis 2021); proactive messaging (Schirmer et al. 2020)	Peer-to-peer knowledge sharing	Farmers exchange locally relevant advice and best practices through peer platforms with offline accessibility for rural areas	Farmers, experts, businesses, translators
		Data coopetition; proactive messaging; insight as offering (Schirmer et al. 2020)	Data coopetition network	Farmers share anonymised data to benchmark performance and generate insights while maintaining data sovereignty and competitive position	Farmers, agribusinesses, data verification providers

Note: The 'BM pattern (references)' column enumerates the identified BM patterns that were used as building blocks in designing the fair BM configuration of each fair BM prototype and the source publications of the patterns.

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the exchange of sufficient and reliable information (DR8), also with procedural fairness mechanisms. **Transparency branding** enables stakeholders to build consumer trust by publicly sharing comprehensive information about environmental impacts, production processes and even challenges or failures in sustainability transitions. **Consumer engagement platform** uses digital tools to provide consumers detailed insights into environmental impacts of their purchases while offering personalised recommendations for sustainable alternatives. **Peer-to-peer knowledge sharing** and **data coopetition network** address information exchange among value chain actors. The former involves platforms where farmers exchange practical advice, best practices and real-time updates with offline capabilities to ensure rural accessibility, whereas a data coopetition network involves collaborative data sharing for benchmarking while maintaining individual data sovereignty. These prototypes share a focus on making decision-relevant information accessible to actors who currently lack it, with procedural fairness mechanisms embedded in value capture and delivery.

4.3 | Demonstration and Evaluation Results

A synthesis of the demonstration and evaluation results across the five case studies is presented in Table 5. The value chain stakeholders have validated fair BM prototypes by assessing the relevance, plausibility and possible implementation challenges of the fair BM implementation when implementing CSA.

Eleven of the 13 prototypes appeared to be highly relevant across all case studies, confirming that stakeholders perceive fairness as key for CSA implementation. Plausibility, however, varied across prototypes and case studies. Low plausibility scores reveal a readiness gap that indicates stakeholders' uncertainty about mobilising their capabilities either due to unfamiliarity with comparable approaches or coordination complexity. Thus, fair BMs with low plausibility scores only suggest that required capabilities, infrastructure or institutional changes are currently not available for successful implementation.

Plausibility for prototypes that address distributive fairness depended on existing institutional arrangements. **Pay-for-success** appears highly plausible only in Denmark where industry initiatives (e.g., Danish Crown's Climate Track Program) provide the necessary measurement architecture. In Germany and the Netherlands, stakeholders assessed plausibility as low, citing the difficulty of accurately measuring CSA outcomes and attributing impacts to specific CSA practice rather than to external factors such as weather. Regarding **outcomes-based pricing** stakeholders also highlighted the difficulty of isolating technology impacts from environmental impacts. The r**isk-sharing fund** required strong oversight mechanisms (for Dutch participants), retailer involvement and clear risk demarcation (for Danish stakeholders) to be seen as plausible. **Consumer-driven fair pricing** was highly plausible in Germany, where stakeholders identified no major challenges. The **crowdfarming** model was mid-level plausible in Spain, where consumer unwillingness to prepay for agricultural products conflicted with the model's core value capture mechanism. The **shared resource pool** model was challenged by the preferences for individual ownership and perceived coordination complexities outweighed potential equipment sharing benefits.

Plausibility for prototypes that address procedural fairness was shaped by cultural norms and existing collaborative practices. **Bio-districts** were highly plausible in Germany, where local communities and state government support territorial and place-based sustainability initiatives such as Öko-Modelregionen (eco model regions). **Retail choice editing** shows low plausibility in both Denmark and the Netherlands, where consumer acceptance, lack of clear revenue model for retailers adopting choice editing and regulatory barriers were raised as critical. **Open source for CSA** plausibility was evaluated as low because of resource constraints and uncertainty about returns on investments.

Plausibility for prototypes that address informational fairness is mainly low due to implementation challenges related to trust, credible certification, data governance and resource requirements. **Transparency branding** revealed challenges around consumer awareness, trust in certification bodies and greenwashing practices. The consumer awareness challenges reflect difficulties in communicating credible information about sustainability benefits, whereas trust in certification bodies and verification systems remains weak. Pricing obstacles mirror broader market dynamics where sustainability claims depend on consumers' willingness to pay premiums. **Consumer engagement platform** was mid plausible in Lithuania, where stakeholders flagged difficulties related to certification processes. **Peer-to-peer knowledge sharing**, in contrast, was highly plausible in Spain where stakeholders assessed mechanisms to engage and facilitate knowledge exchange among experienced and newer farmers as promising and useful tools for youth engagement in agriculture but was challenged by significant barriers in Lithuania including lack of organising bodies, digital literacy gaps and cultural resistance among farmers to seek external advice, which stakeholders attributed to norms of self-reliance in farming communities in Lithuania. **Data coopetition network** has low plausibility in Germany despite existing data-sharing infrastructure revealing challenges around data security, saturation of existing platforms, extra effort needed from farmers on data management and the perception that retailers and processing companies may not be interested in transparency.

In summary, the results show how fairness can be operationalised in BMI. Stakeholder-identified fairness concerns were translated into eight DRs for BMI, generating 13 fair BM prototypes that address distributive, procedural and informational fairness. Stakeholders evaluated 11 fair BM prototypes as highly relevant, but plausibility remained low where contextual conditions challenge implementation. For prototypes addressing distributive fairness, results highlight the crucial role of institutional frameworks (through public policies, subsidies and tools for measuring CSA outcomes), consumer unwillingness to (pre)pay and adopt platforms and governance structures as critical factors in enabling fair BM implementation. For prototypes addressing procedural fairness, plausibility depended on improved coordination among stakeholders and whether more powerful actors (e.g., retailers) were willing to change existing strategies. For prototypes addressing informational fairness, results illustrate that trust capacities and cultural openness to

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TABLE 5 | Comprehensive results of the evaluation.

Fair BM prototypes	Case study	Relevance	Plausibility	Implementation challenges	Validation summary
Pay-for-success	Denmark	High	High	Trade-offs between climate-smart metrics and other sustainability indicators; coordination requirements	Existing industry cooperation and measurement infrastructure support implementation
	Germany	High	Low	Measurement complexities; attribution challenges; farmer planning security concerns	Measurement accuracy concerns and regulatory uncertainty limit plausibility
	Netherlands	High	Low	Uncontrollable variables (weather, markets), causality attribution difficulties	Inability to isolate farming practice impacts challenges implementation
Crowdfarming	Spain	Mid	Mid	Consumer unwillingness to prepay, limited product variety; seasonal constraints	Cultural barriers to prepayment models reduce adoption potential
Consumer-driven fair pricing	Germany	High	High	Consumer engagement requirements; retailer participation challenges	Transparency potential recognised with minimal barriers
Risk-sharing fund	Denmark	High	Low	Retail chain dependency; consumer reluctance for sustainable meat premiums	Funding mechanisms and consumer support remain limit plausibility
	Netherlands	High	Low	Fund governance complexities; compensation criteria disputes; entrepreneurship concerns	Concerns about farmer risk-taking incentives limit plausibility.
Bio-districts	Germany	High	High	Local political dependencies; consumer education requirements	Existing policy frameworks provide strong foundation for implementation
Retail choice editing	Denmark	High	Low	ESG framework immaturity; consumer confusion about reduced choices	Concerns about regulatory and consumer acceptance barriers
	Netherlands	High	Low	Retailer revenue model conflicts; transparency implementation challenges	Retailer resistance to limiting choices challenges plausibility
Open source	Lithuania	High	Low	Funding constraints; human resource limitations; platform maintenance costs	Resource constraints and sustainability concerns challenge plausibility
Outcomes-based pricing	Netherlands	High	Low	Causality measurement in complex agricultural systems; risk allocation disputes	Measurement complexity and risk distribution challenge plausibility
Shared resource pool	Spain	Mid	Mid	Cultural individualism; coordination difficulties; equipment scheduling conflicts	Cooperation challenges and logistics complexity limit plausibility
Transparency branding	Lithuania	High	Mid	Trust deficits in certification bodies; consumer education gaps; greenwashing concerns	Mixed views about certification reliability and consumer response challenge plausibility
Consumer engagement platform	Lithuania	High	Mid	Resource constraints for development; consumer technology adoption barriers	Investment feasibility and user adoption challenge plausibility

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TABLE 5 | (Continued)

Fair BM prototypes	Case study	Relevance	Plausibility	Implementation challenges	Validation summary
Peer-to-peer knowledge sharing	Lithuania	High	Mid	Funding and human resource constraints; platform sustainability challenges due to digital literacy gaps; cultural resistance to asking for help	Strong concept recognition but implementation barriers persist
	Spain	High	High	Participatory platform development requirements; ongoing facilitation needs	High implementation potential with proper support. Model was seen as promising for youth engagement in agriculture
Data coopetition network	Germany	High	Low	Existing platform redundancy; data security concerns; competitive sensitivities	Existing data security issues challenges plausibility

Note: Relevance scores reflect prototypes perceived importance to address fairness concerns in the case studies; plausibility reflects perceived likelihood of successful implementation within current contexts. Relevance and plausibility scores represent group-level consensus reached through collective mapping as part of the evaluation activity.

collective action determined plausibility. These findings provide insights into what conditions are critical for the implementation of fair BMs for CSA.

5 | Discussion, Conclusion and Implications

This study posed the question: *How can fair BMs for climate-smart agriculture be defined, designed and developed?* Using a design science research approach across five European case studies, the study answers the research question by conceptualising and operationalising fairness in BMI. The study makes several contributions, including to theory, methods and policy and management practice.

5.1 | Theoretical Contributions

The study defines a *fair BM as a BM in which the design of the value proposition, value creation and delivery and value capture* enables equitable value distribution, inclusive processes, transparency of information and respectful relationships among stakeholders. This conceptualisation contributes to BMI literature by positioning fair BMs as a distinct concept that operationalises fairness as the central design lens, thereby distinguishing BMs that target sustainability (Schaltegger et al. 2016) and regenerative (Konietzko et al. 2023) outcomes from those that prioritise fairness. This distinction matters strategically because BMs can be perceived as unfair even when they deliver sustainable or regenerative outcomes and can undermine stakeholder trust and long-term commitment. By explicitly addressing stakeholders' (subjective) fairness concerns, fair BMs can build collaborative relationships necessary for sustained implementation of sustainable and regenerative practices like CSA.

The study contributes to BMI and change management research where existing BMI frameworks fail to address the complex multi-stakeholder change processes required for operationalisation of fairness into BMs. Likewise, literature on stakeholder-oriented sustainable BMs (e.g., Evans et al. 2017; Freudenreich et al. 2019) do not specify how stakeholder interests particularly regarding fairness are captured and translated into BM configurations. This study integrates change management into the BMI framework (Figure 1) to ensure that fairness is not retrofitted but built into the BMI from initiation to implementation. The integrated framework is specifically relevant for agri-food BMI that faces not only external barriers, such as regulatory uncertainty and market conditions, but also actor-level barriers rooted in change resistance, risk aversion and limited experience with collaborative innovation (Ulvenblad et al. 2018).

5.2 | Empirical Findings and Implications for Fair BMI in Agri-Food

Across the five case studies, the design science research approach generated DRs for fair BMI and 13 fair BM prototypes that illustrate how fairness can be operationalised in BMI. The eight DRs show that addressing fairness through BMI in European agri-food value chains when implementing CSA requires simultaneous attention to distributive (DR1, DR2 and DR5), procedural

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(DR3 and DR4) and informational fairness (DR6, DR7 and DR8). The absence of interpersonal fairness constitutes a substantive finding about the fairness priorities of stakeholders in the cases studied. This is consistent with Gudbrandsdottir et al. (2021), who argue that the prominence of specific fairness dimensions varies across food systems depending on structural characteristics such as power asymmetries and profit distribution; which fairness concern surfaces is therefore context dependent rather than universal. Additionally, this finding may possibly reflect contexts where formal contracts and market mechanisms mediate most interactions, making interpersonal dynamics less visible as a source of unfairness. However, this absence may also reflect a methodological limitation. The co-creation workshop-based methodology, with its group-based and canvas-mediated exercise, may have reinforced this absence as stakeholders are unlikely to raise interpersonal concerns in workshop settings where actors these concerns relate to are present.

The multidimensional nature of fairness has direct influence on fair BM design. Stakeholders disengage when they perceive they are inadequately rewarded, excluded or poorly informed and not well treated. Addressing these fairness perceptions through BM design means that each fairness dimension engages different actors across the value chain, a pattern that the 13 fair BM prototypes clearly demonstrate. For instance, pay-for-success and risk-sharing fund depend on investors, certifiers and consumers, while Peer-to-Peer Knowledge sharing depends on technology providers and experts. The farmer remains the focal actor in every prototype, but no prototype functions through farmer action alone. Fair BMs by design are therefore structurally dependent on coordinated commitment among value chain actors.

Whether such stakeholder commitment can be mobilised, however, depends on the prototype's institutional requirements and the existing institutional conditions in each context. Across cases, prototypes were assessed as plausible where stakeholders could observe or had experienced the underlying mechanisms in practice and with low plausibility where such precedents were absent. Pay-for-success, for instance, was highly plausible in Denmark where stakeholders pointed to existing industry measurement infrastructure systems (e.g., the Danish Crown Climate Track Program) as supportive of outcome-based models, yet assessed with low plausibility in Germany and the Netherlands where stakeholders cited difficulty of accurately measuring CSA outcomes and attributing impact to specific practices. This low plausibility reflects stakeholders' limited experience with comparable measurement and attribution mechanisms in their context rather than a definitive absence of institutional capacity. By measurement infrastructure, we mean the technical, economic and institutional capacity enabling standardised measurement, verification and reporting of CSA outcome. This infrastructure is not exogenous: It can be built through coordinated investment, policy support and industry collaboration. Low plausibility in this case therefore may also suggest that stakeholders have not yet observed or experienced such coordination in their context, not that it cannot be achieved. Bio-districts was highly plausible in Germany, where stakeholders identified state-supported territorial initiatives, such as the so-called Öko-Modelregionen (in English: organic model regions), providing an existing governance structure for place-based collective action. Retail choice editing showed low plausibility in Denmark and the Netherlands, where stakeholders raised concerns about consumer acceptance, retailer revenue model conflicts and regulatory barriers. Prototypes that stakeholders assessed as highly plausible are those where the enabling mechanisms are already observable in practice, suggesting stakeholders are more likely to commit to their implementation. Where plausibility is low, even highly relevant prototypes remain difficult to implement because the enabling conditions stakeholders lack experience with must first be built. This demands greater coordination effort and carries higher risk of disengagement. These implementation challenges suggest that fair BM implementation must be tailored to contextual conditions and supported by change management activities that, as the integrated framework suggests, progressively build the capabilities, relationships and infrastructures needed to sustain fair BMs over time.

Applying the integrated framework showed the practical challenges of multi-stakeholder co-creation. Multi-stakeholder co-creation in agri-food value chains is challenged by power asymmetries among actors with unequal bargaining positions, by the difficulty of sustaining engagement when short-term benefits are limited and by the substantial coordination effort required to manage participatory processes (Vries et al. 2024). In our case studies, the time, expertise and coordination effort required to facilitate and conduct the co-creation workshops were substantial, and sustaining stakeholder engagement across design science research activities was challenging. These challenges can shape whose fairness concerns surface, how they are prioritised by stakeholders and the overall outcome of the process, which suggest that the framework's effectiveness depends on how co-creation processes are structured and managed in practice.

Taken together, these findings advance our understanding of fair BMI in agri-food by specifying which fairness concerns stakeholders prioritise, which actors those concerns implicate and what conditions make a fair BM implementable, questions that agri-food BM research (Galgo et al. 2025; Isakhanyan et al. 2024; Miranda et al. 2023) has acknowledged as relevant but has not yet addressed.

5.3 | Implications for Design Science Research

Along recent studies advancing design science for multi-actor innovation processes (Smulders et al. 2026), the study demonstrates a co-creation-based design science research process that translates stakeholder-identified fairness concerns into DRs and pattern-based prototypes. This approach builds on advances in BMI literature (Lüdeke-Freund et al. 2018; Remane et al. 2017) and aligns with design science principles (Seckler et al. 2021) while extending both through stakeholder co-creation focusing on fairness.

5.4 | Management and Policy Implications

The fair BM prototypes offer templates that can enable multi-stakeholder collaboration and commitment. For business leaders, the study reveals fairness in agri-food value chains can be

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strategically pursued through fair BMI. Beyond the portfolio of fair BMs, findings from the validation process provide diagnostic insights on what factors are required for fair BM implementation. For policymakers, our findings suggest the creation of supportive and stable regulatory environments that facilitate fair BMI and their scaling. Specific recommendations include developing measurement infrastructure standards that support outcome-based BMs like pay-for-success model, establishing regulatory frameworks that enable innovative value chain arrangements while protecting vulnerable stakeholders, creating incentive structures that reward fairness and environmental outcomes and investing in trust-building institutions that can serve as credible intermediaries in fair BMs.

As climate change pressures intensify and social inequalities deepen, the imperative for fair approaches to sustainable agricultural transitions is urgent. Overall, the study provides an integrated framework and empirically grounded insights that are useful in defining, designing and developing fair BMs for CSA. The central challenge lies not in whether fair BMs can be designed together but in mobilising sufficient stakeholder commitment in building enabling conditions necessary for fair BM implementation while confronting systemic barriers that sustain unfair systems.

5.5 | Limitations and Future Research Directions

The European agri-food context focus limits insights about application in different institutional environments. Future research could focus on other context where different fairness challenges and priorities, such as interpersonal fairness, may be more pronounced. The workshop-based methodology may have limited the surfacing of interpersonal fairness concerns. Individual interviews or confidential reporting mechanisms may better account for interpersonal dynamics, and contexts with more pronounced relational asymmetries may yield different DRs. Participant continuity across the workshop rounds was limited by stakeholder availability; future research applying multi-round co-creation should aim to ensure a maximum of continuity, for instance, through committed stakeholder panels. As the study focused on the conceptualisation, design and the development of fair BMs and the validation methodology focused on perceived relevance and plausibility rather than actual implementation outcomes, the actual impact of the prototypes requires further investigation. Future research could also examine how fair BMs are best implemented, that is, whether they can be implemented alongside existing sustainable and regenerative BMs and under what conditions plurality yields better fairness and sustainability outcomes.

Author Contributions

Christopher Jr. Galgo: conceptualisation, investigation, writing – original draft, methodology, validation, visualisation, writing – review and editing, software, formal analysis, project administration, data curation, resources, supervision. **Gohar Isakhanyan:** conceptualisation, funding acquisition, writing – original draft, methodology, visualisation, writing – review and editing, project administration, supervision, resources. **Jos Bijman:** supervision, resources, conceptualisation, methodology, writing – original draft, writing – review and editing, validation. **Verena Otter:** conceptualisation, supervision, resources, writing – original draft, writing – review and editing. **Katja Logatcheva:** conceptualisation, writing – original draft, writing – review and editing. **Michiel van Galen:** conceptualisation, writing – original draft, validation, writing – review and editing. **Marilena Gemtou:** conceptualisation, funding acquisition, writing – review and editing, supervision. **Liselotte Puggaard:** investigation, conceptualisation, writing – review and editing, validation. **Magdalena Nertinger:** investigation, conceptualisation, writing – review and editing, validation. **Jon Bienzobas Adrián:** conceptualisation, investigation, writing – review and editing, validation.

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Conflicts of Interest

The authors declare no conflicts of interest.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** Second co-creation workshop agenda. **Data S2:** Third co-creation workshop agenda. **Data S3:** Fair business model prototypes for climate-smart agriculture.

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Appendix A: Deductive Qualitative Analysis of the Stakeholder-Identified Changes

Fairness dimension	Stakeholder-identified changes (*UC and actors*)	Design requirements
Distributive (Netherlands, Denmark, Germany)	• Reward growers for extra costs, risks and for a sustainable produce/fair distribution of margins (UC Netherlands: across all actors) • Incentives with biological processes (UC Denmark: policymakers) • Fair subsidies also for small farms (UC Germany: politics)	**DR1: Fair incentives for farmers.** This theme emphasises changes that focus on ensuring that farmers receive adequate financial support, incentives and recognition for implementing climate-smart agriculture practices
Distributive (Lithuania, Netherlands, Germany, Denmark)	• Changing the distribution of financial support to climate-friendly practices and promotion of organic farming (UC Lithuania: farmers, policymakers, grain buyer) • Trade and retail already are earning money with PlanetProof, now its time for the growers to get their share (UC Netherlands: farmer) • Increase in planning security for all actors (UC Germany: politics, all actors of the value chain)	**DR2: Equitable distribution of benefits across the value chain.** This theme emphasises changes that ensure that value is distributed equitably among different actors in the agri-food system, particularly for farmers' sustainability contribution through CSA practice implementation
Procedural (Netherlands, Germany, Spain, Denmark)	• Less regulation and more facilitation towards sustainable produce (UC Netherlands: value chain) • Changing the current market regulations and practices (UC Germany: politics) • Removal of taxation on organic and staple food products (UC Germany: politics) • Same rules and regulations for all food imports (UC Germany: politics) • Flexibilisation of hygienic sanitary requirements in processing (UC Spain: public administration, advisor, farmer, processor) • Public procurement (to increase demand for organic and fair products) (UC Germany: politics) • There is lack of incentive from the consumer to buy more climate-friendly produce (UC Denmark: societal trend and consumers) • Commercialisation of climate-smart products—needs to involve the retail part, as it is their responsibility to showcase the products for the consumers. We need to approach this backwards (UC Denmark: retailers, consumers, policymakers, farmers)	**DR3: Facilitation (not regulation) towards sustainability.** This theme emphasises facilitation towards CSA. It includes changes to public procurement, subsidies and regulations to create a more enabling environment for climate-smart and fair agri-food systems
Procedural (Spain, Lithuania, Denmark)	• Promotion of consumer associationism and short marketing channels (UC Spain: farmer, public administration, consumer, processor, commercialiser) • Increasing cooperation between small and medium-sized farmers to improve their competitiveness (UC Lithuania: farmers, tech provider, policymaker) • The consumer and retailer need to be closer to the manufacturer. Willingness to create a greater commitment (UC Denmark: societal trend and consumers)	**DR4: Engagement and collaboration among stakeholders.** This theme highlights the need for greater stakeholder engagement and inclusive decision-making processes. This theme focuses on ensuring that policies and market strategies are co-created with input from a diverse set of actors, including farmers, retailers, consumers and policymakers
Procedural (Lithuania, Spain)	• Increasing farmers' access to innovative technologies (UC Lithuania: policymaker, tech provider) • Use of more efficient machinery (UC Spain: farmer, advisor, researcher, processor, commercialiser) • The cultivation technology and technical solutions must be chosen (UC Lithuania: farmers, grain buyer, tech provider) • Transition towards more sustainable agriculture practices (UC Spain: farmer, advisor, researcher, processor, commercialiser)	**DR5: Improve access to technology.** The changes under this theme focus on providing equitable access to technology and innovation, particularly for small-scale farmers. It also includes the development of technological solutions that support CSA and improve farmer's competitiveness
Informational (Netherlands, Denmark)	• Make the value chain transparent (UC Netherlands: across all actors) • Optimisation, requirements, structure and practicality of regulations must change. Policymakers and consumers need more knowledge about how production is done in a more practical way (UC Denmark: policymakers) • The data and impact of CSAs become clearer when presented in an ESG context revealing where fairness exists (UC Denmark: regulations, policymakers)	**DR6: Transparency among value chain actors.** This theme encompasses changes relating to increase transparency in the value chain and provide accurate, reliable information to consumers and stakeholders. This theme focuses on ensuring that actors have access to high-quality information regarding sustainable practices, regulations and value distribution

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Fairness dimension	Stakeholder-identified changes (*UC and actors*)	Design requirements
Informational (Spain, Germany, Denmark, Lithuania)	• Awareness-raising work with consumers about the importance and impact of buying local and sustainably produced products (UC Spain: farmer, public administration, consumer, processor) • Consumer education (honest advertisement and information, campaigns, fair education and marketing, communicate added value in a fair way) (UC Germany: schools, politics, Öko-Modelregionen, dairy) • Holistic definition of fairness and climate-smart (UC Germany: media, science including farmers and practitioners) • Increase demand for organic products (UC Germany: politics, dairy, farmer association, retail [advertisement], farmers, media) • Increasing consumer awareness of and trust in products bearing the organic label (UC Lithuania: farmers, technology provider, grain buyer) • Origin labelling (UC Germany: actors, retail, gastronomy, tourism)	**DR7: Consumer awareness and knowledge about sustainable choices.** This theme focuses on educating consumers about the benefits of sustainable and fair practices, as well as providing clear and accessible information to influence purchasing behaviour. It aims to create a more informed consumer base that supports sustainability through purchasing decisions
Informational (Spain, Germany, Denmark)	• Research/increased knowledge of local fruit varieties for their promotion and improvement of yields (e.g., pollination behaviour) (UC Spain: public administration, advisor, researcher, farmer) • Holistic definition of fairness and climate-smart (UC Germany: media, science including farmers and practitioners) • The data and impact of CSAs become clearer when presented in an ESG context revealing where fairness exists (UC Denmark: regulations, policymakers)	**DR8: Exchange of sufficient and reliable information.** This theme focuses on ensuring access to trustworthy, timely and comprehensive information. It includes efforts to enhance the flow of accurate data on sustainable practices, fair value distribution and regulatory standards

Note: Of the 33 stakeholder-identified changes, six fall under distributive fairness (Netherlands (2), Denmark (1), Germany (2), Lithuania (1)). Fifteen fall under procedural fairness (Netherlands (1), Denmark (3), Germany (4), Lithuania (3), Spain (4)) and twelve for informational fairness (Netherlands (1), Denmark (3), Germany (5), Lithuania (1), Spain (2)). No change was classified under interpersonal fairness in any case study.

Appendix B: Sources of BM Patterns

	Name of publication	Journal	Year of publication	Number of patterns
1	Business Models for Sustainable Food Systems: A Typology Based on a Literature Review (Donner and de Vries 2023)	*Frontiers in Sustainable Food System*	2023	9
2	A Systematic Review of the Literature on Agri-Food Business Models: Critical Review and Research Agenda (Miranda et al. 2023)	*British Food Journal*	2023	16
3	A Typology of Sustainable Circular Business Models With Applications in the Bioeconomy (Keyser and Mathijs 2023)	*Frontiers in Sustainable Food Systems*	2023	6
4	Unpacking Sustainable Business Models in the Swedish Agricultural Sector – The Challenges of Technological, Social and Organisational Innovation (Barth et al. 2021)	*Journal of Cleaner Production*	2021	8
5	The Sustainable Business Model Database: 92 Patterns That Enable Sustainability in Business Model Innovation (Schroedel 2023)	*Sustainability*	2023	92
6	Open Business Models and Close-Loop Value Chains (Kortmann and Piller 2016)	*California Management Review*	2016	9
7	32 Ways to Innovate Business Models Through Data: Emerging Data-Driven Solution Business Model Patterns From a Study of 471 Late-Stage Data-Driven Start-Ups (Schirmer et al. 2020)	*HICSS Proceedings*	2020	30
8	Emerging Innovation Patterns in Digital Agriculture: A Study of 198 Digital Solutions From 116 Startups (Schirmer et al. 2021)	*HICSS Proceedings*	2020	19
9	Digital Sustainable Business Models: Using Digital Technology to Integrate Ecological Sustainability Into the Core of the Business Model (Böttcher et al. 2024)	*Information Systems Journal*	2021	4

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	Name of publication	Journal	Year of publication	Number of patterns
10	Best Business Models for Fast-Moving Consumer Goods Sector: Patterns for Innovation (Schroedel 2024)	*Sustainability*	2024	12
11	From Theory to Practice: Systematising and Testing Business Model Archetypes for Circular Economy (Pieroni et al. 2020)	*Resources, Conservation and Recycling*	2020	20
12	Three Circular Business Models That Extend Product Value and Their Contribution to Resource Efficiency (Whalen 2019)	*Journal of Cleaner Production*	2019	3
13	Archetypes of Business Models for Circular Economy: A Classification Approach and Value Perspective (Woldeyes et al. 2023)	*Sustainable Design and Manufacturing*	2023	23
14	Leveraging Industry 4.0 – A Business Model Pattern Framework (Weking et al. 2020)	*International Journal of Production Economics*	2020	10
15	An Overview of Sustainable Business Models for Innovation in Swedish Agri-Food Production (Ulvenblad et al. 2019)	*Journal of Integrative Environmental Sciences*	2019	8
16	A Literature Review and Practice Review to Develop Sustainable Business Model Archetypes (Bocken et al. 2014)	*Journal of Cleaner Production*	2014	52
17	Business Model Patterns in the Sharing Economy (Curtis 2021)	*Sustainable Production and Consumption*	2021	8
18	A New Circular Business Model Typology to Creating Value From Agro-Waste (Donner et al. 2020)	*Sustainable Production and Consumption*	2021	6
19	Business Models for the Internet of Things (Kralewski 2016)	*Information Systems*	2016	20
20	Digital Sustainable Business Models: Using Digital Technology to Integrate Ecological Sustainability Into the Core of Business Models (Böttcher et al. 2024)	*Information Systems Journal*	2024	4
21	Discovering Business Models of Data Marketplaces (Fruhwirth et al. 2020)	*HICSS Proceedings*	2020	4
22	A Pattern-Based Tool to Support Companies in Sustainability-Oriented Business Strategies: Grounded Theory Research in Agribusiness (Latino et al. 2024)	*Corporate Social Responsibility and Environmental Management*	2024	14
23	The Sustainable Business Model Pattern Taxonomy – 45 Patterns to Support Sustainability-Oriented Business Model Innovation (Lüdeke-Freund et al. 2018)	*Sustainable Production and Consumption*	2018	45
24	The Business Model Pattern Database—A Tool for Systematic Business Model Innovation (Remane et al. 2017)	*International Journal of Innovation Management*	2022	49
25	Green Entrepreneurship and Business Models: Delivering Green Technology Business Model Archetypes (Trapp and Kanbach 2021)	*Journal of Cleaner Production*	2021	12

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